

Vehicle Trends and Infrastructure Planning

Project Title: Vehicle Trends and Infrastructure Planning

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Description:

This project analyzes vehicle population trends and growth opportunities in private and commercial markets in Maharashtra, India. The goal is to provide actionable insights into sustainable transport strategies and guide policy planning to address challenges like pollution, congestion, and infrastructure strain.

Using Python-based analysis and visualizations, the project highlights regional disparities, predicts future trends, and suggests solutions to optimize urban mobility, including public transportation, electric vehicles (EVs), and green infrastructure.

Input Files:

1. **Vehicle Population Dataset** (Excel file):
 - Contains information about vehicle categories, population, and trends across regions.
 - Key columns:
 - Year: Year of data collection.
 - Region: Region in Maharashtra.
 - Private Vehicles TOTAL: Total private vehicles.
 - Commercial vehicles TOTAL: Total commercial vehicles.
 - Vehicle-specific categories (e.g., Motorcycles, Scooters, etc.).
 2. **Environmental Data** (Included in Research Document):
 - LSI/MSI and pollution levels in different cities.
 - Regional vehicle registration and pollution load data.
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Output Files:

1. Visualizations:

- Bar charts for vehicle trends by region.
- Line graphs for private vs. commercial vehicle growth over time.
- Heatmaps showcasing correlations between vehicle categories.
- Growth rate comparisons for top regions.

2. Insights:

- Regional disparities in vehicle growth.
- Recommendations for public transport and EV adoption.
- Identification of high-demand areas for policy planning.

3. Predictive Results:

- Forecast of private vehicle growth based on current trends using regression models.
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Main Code File:

1. Google Colab Notebook (SourceCode.ipynb):

- Processes the vehicle population dataset.
 - Cleans and converts data for analysis.
 - Performs Exploratory Data Analysis (EDA) using:
 - Matplotlib
 - Seaborn
 - Predictive modeling using Linear Regression from sklearn.
 - Combines data cleaning, visualization, and prediction tasks.
 - Interactive for running analysis in Google Colab.
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Key Steps in the Code:

1. Data Cleaning:

- Remove missing or invalid values.
- Convert object columns to numeric types.
- Standardize column names.

2. Exploratory Data Analysis (EDA):

- Plot trends for vehicle populations by region.

- Analyze correlations between vehicle categories using heatmaps.

3. **Trend Analysis:**

- Examine growth rates for private and commercial vehicles over time.
- Identify the top regions with the highest growth.

4. **Predictive Modeling:**

- Train and evaluate a Linear Regression model to predict private vehicle growth.
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Insights from Analysis:

1. **Regional Disparities:**

- Pune leads in private vehicle growth (467%), while Nashik shows a significant commercial vehicle rise (250%).

2. **Key Challenges:**

- High traffic congestion in urban areas like Pune and Mumbai.
- Rising emissions due to the dominance of private vehicles.

3. **Recommendations:**

- Expand public transport networks.
 - Transition to electric vehicles with government subsidies.
 - Invest in green infrastructure like pedestrian walkways and cycling lanes
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How to Run the Project:

1. **Setup:**

- Install required libraries: pandas, numpy, matplotlib, seaborn, sklearn.
- Ensure the dataset file is available in the project directory.

2. **Run the Analysis:**

- Execute the Python script analysis.py or open ipynb (SourceCode.ipynb) file in Google Colab.

3. **Review Outputs:**

- View generated graphs and summary insights.

- Check predictive results for future vehicle trends.
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References:

- <https://www.hindustantimes.com/cities/mumbai-news/more-than-25-lakh-vehicles-registered-in-maharashtra-in-2023-7-91-pc-rise-in-one-year-101704113134907.html>
- <https://www.statista.com/statistics/831009/total-number-of-commercial-vehicles-in-maharashtra-india/#:~:text=Services-.About%20Statista.within%20the%20segment%20that%20year.>
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